Boosting LLM's Molecular Structure Elucidation with Knowledge-Enhanced Tree Search Reasoning

The Challenge: LLMs in Chemistry

Molecular structure elucidation is the process of determining a molecule's structure from spectral data (like NMR and IR spectroscopy). This is a critical but complex task in chemistry, often taking skilled professionals 10-15 minutes per molecule.

While Large Language Models (LLMs) are great at complex reasoning, they face two major hurdles in this domain:

1. **Limited Chemical Knowledge:** LLMs lack a comprehensive understanding of the vast chemical structure space. For example, they might see data for a **thiophene ring** (4 carbons, 1 sulfur) and incorrectly identify it as a common **benzene ring** (6 carbons).
2. **Poor Self-Evaluation:** LLMs cannot accurately judge their proposed solutions' quality, which is essential for guiding reasoning processes like tree search. (self-affirmation bias)

The Solution: The K-MSE Framework

The paper introduces **K-MSE (Knowledge-enhanced reasoning framework for Molecular Structure Elucidation)**, a plugin that significantly improves an LLM's ability to solve these molecular puzzles.

It is built on three core ideas:

1. **A Molecular Substructure Knowledge Base** to supplement the LLM's limited chemical knowledge.
2. **A Specialized Molecule-Spectrum Scorer** to act as an accurate reward model, guiding the reasoning process, is also used during the retrieval part from the knowledge base.
3. **A Monte Carlo Tree Search (MCTS) Framework** to enable effective, iterative reasoning and self-correction at test time.

Molecular Substructure Knowledge Base

Purpose: To expand the LLM's coverage of the chemical structure space with domain-specific knowledge.

Construction Process

1. **Substructure Collection:** Ring and chain substructures were extracted (extraction process described in Appendix A.1) from over 4 million molecules in the **Moses database**. To ensure a balance of diversity and commonality, only substructures appearing more than 1,000 times were kept, resulting in **593 unique substructures**.
2. **Description Generation:** Generating accurate natural language descriptions for these substructures is key. To do this, an automated pipeline was created:
    - For each substructure (in SMILES format), supplementary information like its formula,

image, and structural details (e.g., ring size, atom types) was extracted using the **RDKit** tool.
    - **GPT-4o-mini** was then prompted with this rich context to generate a concise and accurate textual description for each of the 593 substructures.

This knowledge base provides the LLM with readily accessible information about less common but important molecular fragments during reasoning.

Molecule-Spectrum Scorer

Purpose: To act as a reliable, specialized reward model that accurately evaluates the alignment between a predicted molecule and the given spectral data.

Architecture & Training

This scorer consists of two main parts:

- **Molecule Encoder (**gm**):** Encodes the proposed molecular structure. It takes two inputs:
    - The **molecule graph**, processed by a Graph Isomorphism Network (GIN). (5 layers, dim=300)
    - The **molecule fingerprint** (a combination of Morgan (dim=1024), MACCS (dim=167), and RDK (dim=1024) fingerprints), processed by a Multilayer Perceptron (MLP) (2 layers). (fixed-length vectors giving information about whether a particular substructure is present or not) (2215 dim -> 300 dim)
    - The encoded molecular graph and fingerprint are concatenated, passed through two additional layers of MLP, and ultimately produce a molecular representation with a dimension of 256.
- **Spectrum Encoder (**gs**):** Encodes the NMR spectral data (C-NMR (2500 tokens) and H-NMR (1500 tokens)). It tokenizes features like chemical shifts, splitting patterns (75 tokens), and coupling constants (200 tokens) and processes them through a **Transformer** encoder. (2 layers, dim=256)

Training:

- **Dataset:** The scorer was trained on a dataset of **9,000 molecules** from the **ZINC database**, with their corresponding NMR spectra generated via simulation. An additional 200 molecules were used for validation.
- **Objective:** The model was trained using **NT-Xent contrastive loss** over 100 epochs to learn embeddings where a molecule and its correct spectrum are close together in the embedding space. (tau = 0.07)

MCTS-based Reasoning Framework

Purpose: To integrate the knowledge base and scorer into a dynamic reasoning process that allows the LLM to explore different solutions, reflect on them, and iteratively improve its answer.

The K-MSE Reasoning Flow

1. **Retrieval:** The process starts by using the **spectrum encoder (**gs**)** from the scorer to find the **Top-k** most relevant substructures from the knowledge base based on the input spectrum. These are added to the prompt.
2. **MCTS Iterations:** The framework then performs N_iter iterations of the MCTS algorithm:
    - **Selection:** Chooses the most promising solution path so far using the UCT algorithm. (c=1, epsilon=0.1)
    - **Expansion:** The LLM **critiques** the current solution. To make this critique reliable, it's given not just the text but also the **molecular image and formula**. It then **rewrites** the solution based on the critique, creating a new "child" node. (max 2 child nodes per node)
    - **Evaluation:** The new solution is evaluated by the **specialized scorer**, which calculates a reward score based on how well the predicted molecule matches the spectrum.
    - **Backpropagation:** This reward is propagated back up the tree to update the scores of parent nodes, guiding future selections toward more promising areas.
3. **Final Answer:** After all iterations, the solution with the highest reward is chosen as the final answer.

Experimental Setup

- **Benchmark Dataset:** The **MolPuzzle** benchmark, which contains **216 molecules**.
- **Task:** "Addressing Entire Molecule Puzzles" in a **zero-shot** manner. The model is given IR spectroscopy, C-NMR, H-NMR, and the molecular formula, and must predict the final structure.
- **Base LLMs:** Llama-3.2-11B, GPT-4o-mini, GPT-4o, and GPT-o1.
- **Baseline Methods:**
    - Chain-of-Thought (CoT)
    - Self-Refine (Self-R)
    - Self-Consistency (Self-C)
    - Multi-Agent Debate (MAD)
    - MCT Self-Refine (MCTSr)
- **Evaluation Metrics:**
    - **Fingerprint Tanimoto Similarity (FTS):** Measures chemical similarity.
    - **Formula Accuracy (Formula Acc):** Checks if the predicted formula is correct.
    - **ACC:** Exact structural match (the strictest metric).
- **Hyperparameters:**
    - MCTS iterations (N_iter): When the base models are Llama-3.2-11B, GPT-4o-mini, and GPT-4o, N_iter is set to 8; when the base model is GPT-o1, N_iter is set to 2.
    - Retrieval count (k): For Llama-3.2-11B and GPT-4o-mini, the number of retrieval count k from the knowledge base is set to 1; while for GPT-4o and GPT-o1, k is set to 2
    - To maximize reproducibility, we set the temperature to 0.0 (most conservative) and top_p to 1.0

- Prompts are given on the last page

Experimental Results & Analysis

Main Results

K-MSE significantly outperformed all baseline methods across all models.

- On **GPT-4o**, ACC (exact match accuracy) jumped from 27.8% to **57.8%** (+0.300).
- On **GPT-4o-mini**, ACC rose from 3.7% to **27.3%** (+0.236).
- Some baselines like MAD and MCTSr actually *hurt* performance on weaker models because they relied on the LLM's poor self-correction abilities, which amplified errors.

Ablation Studies: Why K-MSE Works

1. **Specialized Scorer is Crucial:**
   - Compared to an LLM-based scorer, the specialized scorer was far superior, boosting ACC by **0.139**.
   - The scorer's reward showed a **strong positive correlation (0.53)** with the true similarity, while the LLM's score had almost no correlation (0.03), proving the LLM is a biased and unreliable judge.
2. **Scorer is a Better Retriever:** Using the scorer to retrieve knowledge was much more effective than an LLM+BM25 approach, improving ACC by **0.120** and increasing the retrieval hit rate.
3. **Knowledge Base is Universally Helpful:** Adding the knowledge base as a plugin improved performance across **all baseline methods**, demonstrating its flexibility and effectiveness.
4. **Critique Needs Context:** Removing the molecular **image** or **formula** during the critique step caused performance to drop, showing that this extra context is vital for helping the LLM accurately identify flaws in its previous reasoning step.